
RAML Report: Quantum Implicit Neural Representation

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Abstract

Quantum Implicit Neural Representation (QIREN) uses trainable quantum circuits to model signals such as images and audio as continuous functions. Standard classical (i.e., non-quantum) models struggle to capture the fine-grained, high-frequency details those functions can contain. As a solution, the QIREN model was proposed to address these limitations with data reuploading. We assess its expressive power on image representation tasks, and experimentally benchmark it against a classical Multi-Layer Perceptron (MLP) with Random Fourier Features (RFF). We use the PSNR values of the images and conclude that the QIREN outperforms the MLP+RFF architecture.

1 Introduction

Implicit neural representation (INR) parametrizes the implicit function of a signal. It finds application in representation of images, and sounds. Early works[1][2][3][4] have built upon this using Deep Multi Layer Perceptrons (MLPs), but these architectures lack expressivity as they tend to smooth out the high frequencies, resulting in blurry images as outputs. The standard building blocks of INR are deep MLPs. They exhibit a spectral bias: low-frequency components are learned quickly, whereas high-frequency details converge slowly or are smoothed out entirely. Fourier Neural Networks (FNNs) try to fix the “blurry-details” problem by sprinkling sine and cosine terms into the input or hidden layers. FNNs[5] learn the underlying signal in its Fourier representation, allowing to control the frequencies needed to represent the signal. The trigonometric bases let the network follow higher-frequency patterns but each extra frequency needs its own weights. As a result, the total frequency “bandwidth” of an FNN only grows in direct proportion to the number of parameters. For today’s ultra-detailed images, that quickly forces the model to become very wide, which means slower training and a much larger memory footprint.

Following the idea of FNNs, the authors Zhao et al.[6] proposed Quantum Implicit Neural Representation (QIREN), which replaces the first linear layer of a coordinate MLP with a trainable, data-re-uploading quantum circuit. The data re-uploading imitates what Fourier features are able to achieve, it can represent higher frequencies, solving the problem of blurry outputs by capturing more details. The quantum layer expands the coordinate into a set of basis functions whose frequencies grow exponentially with circuit depth, after which a shallow classical MLP predicts the final signal value. All parameters, quantum rotation angles and classical weights, are optimised jointly by gradient descent via a differentiable quantum simulator or hardware back-end.

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In this work, we conduct our independent validation of QIREN method. We first review the literature associated with the classical methods and study their limitations. Then we review QIREN methodology and understand its advantages with the data re-uploading incorporated. Subsequently, we apply QIREN to the tasks of Image Regression and Image Super-Resolution to compare it experimentally with the classical methods.

2 Background and Related Work

Quantum machine learning (QML)[7] builds on the *quantum circuit model of computation*: an n -qubit register spans a 2^n -dimensional Hilbert space, can be initialized in any computational basis state, manipulated by a *finite universal gate set* (e.g., Hadamard, Phase, CNOT, $\pi/8$, or Toffoli), and finally measured in the computational basis. Single-qubit gates act as rotations on the Bloch sphere, while controlled gates such as CNOT create entanglement. Larger algorithms are composed from these primitives. Shor-type algorithms that use the quantum Fourier transform demonstrate *exponential* speedups for factoring and discrete logarithms, while Grover’s search provides a *quadratic* advantage [8], motivating the design and experimentation of quantum algorithms for machine learning.

In the near-term **CQ setting**—where classical data is processed by quantum circuits—inputs are first *encoded*, typically via amplitude or time encodings, into quantum states. This step bridges the bit–qubit divide and remains a key bottleneck in QML pipelines. *Variational quantum circuits* (VQCs) then serve as trainable models, with gate angles as parameters. Output expectations are estimated through multiple measurement shots. More advanced techniques, such as quantum teleportation and error correction, can be employed to preserve quantum information where necessary[9]. The practical QML workflow therefore reads: prepare classical inputs $\rightarrow$ embed in Hilbert space $\rightarrow$ process with a variational circuit drawn from the universal gate set $\rightarrow$ measure repeatedly to obtain classical outputs, with optional use of algorithmic motifs like Shor’s or Grover’s to target larger speed-ups.

Recent theory links *encoding design* to model expressivity. Schuld *et al.* [10] prove that any variational circuit built from alternating encoding layers $S(x) = e^{-ixH}$ and trainable blocks realises a finite Fourier series

$$f(x) = \sum_{\omega \in \Omega} c_{\omega} e^{i\omega x}, \quad (1)$$

whose frequency set Ω is fixed by the spectrum of the encoding Hamiltonian H . Increasing depth or qubit count enlarges Ω , and with a sufficiently rich H the circuit becomes a universal approximator on $L^2[0, 2\pi]$. Their design rules highlight (i) choosing Hamiltonians that deliver the desired frequencies, (ii) repeating encodings to grow Ω , (iii) rescaling inputs to match the circuit’s natural periodicity, (iv) classical preprocessing to enlarge Ω indirectly, and (v) making the measurement observable trainable—while warning that high capacity must still be balanced against over-fitting.

Complementing this analysis, Pérez-Salinas *et al.* [11] introduce *data re-uploading* circuits, showing that a *single qubit*—by alternately encoding the input and applying trainable rotations— can already act as a universal classifier. Practical recommendations include fidelity-based cost functions, L-BFGS-B optimisation on small datasets and mini-batch SGD on larger ones, input chunking for high-dimensional data, and adding entangling gates (e.g. CZ) to widen the learned feature map. Together, these works supply both the mathematical foundation and hands-on recipes for constructing compact yet expressive QML models.

3 Method

In this section, we discuss the classical implicit representations in detail, and then consecutively present the QIREN method as proposed by Zhao *et al.*[6] for image representation. We also discuss the architecture of the model, and what data-reuploading implies in this method, specifically, its effect in increase the high frequency representation in a spectrum.

3.1 Classical Implicit-Representation Networks

Two common INR baselines are a ReLU multilayer perceptron with a Random Fourier Feature (RFF) input mapping and the Sinusoidal Representation Network (SIREN). They can both be expressed as *shallow Fourier Neural Networks* (FNNs). Fourier features have trigonometric functions which capture high frequency signals. As a reference, if x is the input, a one-layer ReLU network with RFF

encoding[12] can be expressed as

$$f(\mathbf{x}) = \sum_{i=1}^D a_i \cdot \cos(\mathbf{w}_i^\top \mathbf{x} + b_i) = \mathbf{a}^\top \mathbf{z}(\mathbf{x}) \quad (2)$$

where

$$\mathbf{x} \in \mathbb{R}^d, \quad \mathbf{w}_i \sim p(\mathbf{w}), \quad b_i \sim \text{Uniform}[0, 2\pi], \quad \mathbf{a} \in \mathbb{R}^D,$$

and the coefficient $\mathbf{a}_i$ are trainable parameters. In the experiments, we will benchmark QIREN against this classical RFF-ReLU model.

3.2 Quantum Implicit Representation Network (QIREN)

Quantum Implicit Representation Network (QIREN) leverages data re-uploading in quantum circuits to enhance frequency representation, potentially outperforming classical models in image regression and super-resolution tasks [6]. reuploading means encoding the input data multiple times into a quantum circuit at different layers.

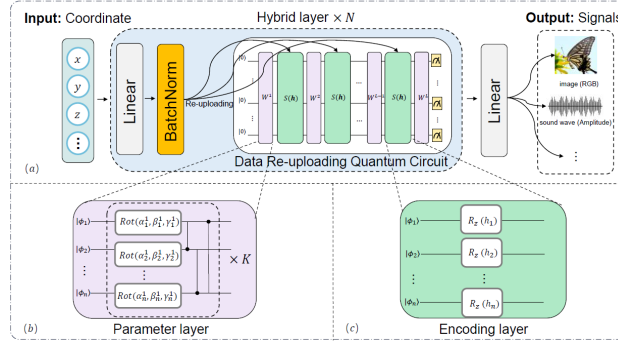


Figure 1: QIREN architecture[6]. The model processes coordinate-based inputs through a hybrid layer consisting of a classical Linear and BatchNorm block followed by a data re-uploading quantum circuit with encoding and parameter layers. This is followed by a final linear layer that produces the output.

Architecture. QIREN replaces the fixed trigonometric front-end of an FNN with a trainable *data re-uploading quantum circuit* (QC). Each of its N hybrid layers performs

$$h_i = \text{BN}(Wx_i + b), \quad (3)$$

$$f_i = [\text{QC}(h_i, O_1), \dots, \text{QC}(h_i, O_{d_f})], \quad (4)$$

where a linear map W expands the input coordinate x_i , batch normalisation stabilises magnitudes, and the QC alternates an encoding block $S(h) = \bigotimes_q e^{-ih_q Z/2}$ with K layers of parameterised single-qubit ROT gates plus entangling CNOTs. When we combine the data reuploading features of QIREN with the linear layers in the architecture described in figure 1, the following mathematical transformation is applied to the inputs:

$$U(\mathbf{x}, \boldsymbol{\theta}) = \prod_{\ell=1}^L \left(\bigotimes_{j=1}^d R_Y(\omega_{\ell j} x_j + \phi_{\ell j}) \right) V(\boldsymbol{\theta}_\ell). \quad (5)$$

where $R_Y(\alpha) = e^{-i\frac{\alpha}{2}Y}$ is a rotation around the Y-axis by angle α , $\mathbf{x} = (x_1, \dots, x_d)$ is the input vector, ω_{ij}, ϕ_{ij} are trainable or fixed random frequencies and phases (as in RFF), $V(\boldsymbol{\theta})$ is a trainable variational unitary (e.g., entangling layer), and L is the number of data re-uploading layers. The input $\mathbf{x}$ is *reuploaded* at each layer ℓ via these parameterized rotations. Stacking the N hybrid layers yields a feature vector $f_i^{(N)}$; a final linear layer maps this to the signal prediction y'_i , and training minimises the mean-squared error over $\{(x_i, y_i)\}$. Equations (6) and (7) show that QIREN's output $f(\mathbf{h})$ can be expressed as a weighted sum of complex exponentials, equivalent to a Fourier series. The

quantum circuit structure defines frequency components via $\Lambda_{\mathbf{K}} - \Lambda_{\mathbf{J}}$, enabling QIREN to capture high-frequency signal features more effectively than classical FNNs.

$$f(\mathbf{h}) = \langle \psi | O | \psi \rangle = \sum_{\mathbf{K}, \mathbf{J}} e^{i(\Lambda_{\mathbf{K}} - \Lambda_{\mathbf{J}}) \cdot \mathbf{h}} \sum_{i, i'} W_{k_1^{(1)}}^{(1)\dagger} W_{k^{(2)} k^{(1)}}^{(2)\dagger} \cdots W_{i k^{(L-1)}}^{(L)\dagger} O_{i, i'} W_{i' j^{(L-1)}}^{(L)} \cdots W_{j^{(2)} j^{(1)}}^{(2)} W_{j_1^{(1)}}^{(1)}. \quad (6)$$

$$f(\mathbf{h}) = \sum_{\mathbf{K}, \mathbf{J}} a_{\mathbf{K}, \mathbf{J}} e^{i(\Lambda_{\mathbf{K}} - \Lambda_{\mathbf{J}}) \cdot \mathbf{h}}, \quad (7)$$

Theoretical guarantees. Let the re-uploading operator S be defined as:

$$S(x) = \exp(-ixH), \quad H = \frac{1}{2} \sum_{q=1}^{d_h} Z^{(q)}. \quad (8)$$

Here, H is a diagonal Hamiltonian in the Pauli- Z basis acting on d_h qubits, and $Z^{(q)}$ denotes the Pauli- Z operator on the q^{th} qubit. The authors Zhao et al. [6] derive the following claims of the QIREN model:

- **Exponential capacity:** Frequency components grow exponentially with the number of re-uploading layers L and qubits d_h , yielding a rich Fourier spectrum.
- **Spectrum densification:** The linear layer W enhances frequency diversity by removing spectral degeneracies.
- **Faster convergence:** Batch normalization improves optimization dynamics, enabling up to $1.5 \times$ faster training.

4 Experiments

In order to test the capabilities of the data re-uploading feature of the QIREN[6], we test its performance on a 64x64 image with the following tasks and the following pre-processing employed;

- **Image Regression.** Reconstructing an image by learning a continuous function that maps spatial coordinates (x,y) to RGB values. all 64x64 pixel locations were converted into coordinate-intensity pairs. Each pixel's (x,y) position was normalized to [-1,1], and the RGB values were normalized to [0,1] as targets.
- **Image Super-Resolution.** Learning from a downsampled image to predict a higher-resolution version by extrapolating RGB values at unseen coordinates.

Both tasks explore the ability of models to learn implicit representations of images based solely on coordinate-pixel pairs. Both models were trained to minimize the Mean Squared Error (MSE) between predicted and true RGB values. Peak Signal to Noise Ratio (PSNR), which expresses the reconstruction error on a decibel scale, is used as the metric for comparing the results from both models.

Both models were trained for 500 epochs with Adam with learning rate of 0.01. The Quantum model uses 7 qbits with data-reuploading layers, while the classical baseline relies on 512 Random Fourier Features. The total number of trainable parameters are 4042 for QIREN and 70000 for the classical model.

4.1 Results

Task	Model	PSNR (dB)
Image Regression	RFF + ReLU	22.49
Image Regression	Quantum Hybrid	33.26
Super-Resolution	RFF + ReLU	20.07
Super-Resolution	Quantum Hybrid	29.83

Table 1: PSNR results for both tasks and models. QIREN model performs significantly better than RFF+ReLU model

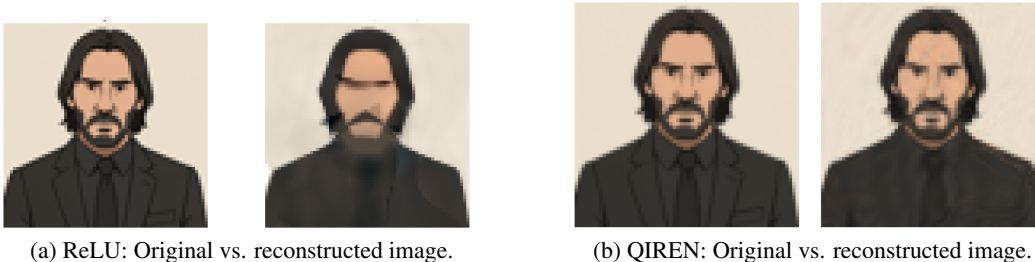


Figure 2: Comparison of Image Fitting: ReLU vs. QIREN. The QIREN-reconstructed image closely resembles the original, preserving fine details, whereas the ReLU-based reconstruction exhibits noticeable feature loss.

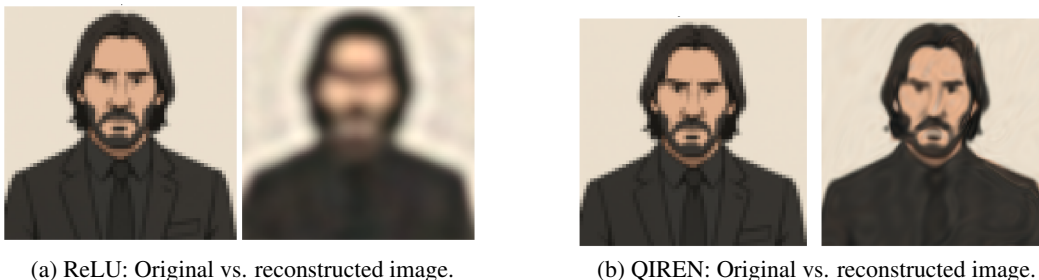


Figure 3: Comparison of Super-Resolution: ReLU vs. QIREN. The ReLU reconstruction suffers from loss of detail, whereas the QIREN output preserves sharper edges and finer textures.

Table 1 reports the experimental results achieved by each model and shows that the QIREN has better image quality by roughly 10 dB over the classical RFF+ReLU baseline in both the image-regression and super-resolution tasks. This can also be visually seen in figures 2 and 3 where the image produced by the classical model is blurry and missing details while the QIREN model closely resembles the original image.

Based on the quantitative scores in Table 1 and the qualitative reconstructions in the figures, three main takeaways emerge:

- The Quantum model performs comparably better with 4042 trainable parameters while the results with RFF + ReLU model is lacking features in the reproduced image.
- Quantum models benefit from data reuploading and entanglement, and may offer advantages in low-data regimes or where classical capacity is limited.
- The PSNR gap between regression and super-resolution is expected due to sparse training in the latter.

5 Conclusion

In this work, we explored QIREN for two image processing tasks: rgb image regression and image super-resolution. To learn the mapping from spatial coordinates to pixel intensities, we compared two concrete models: a purely classical baseline that feeds Random Fourier Feature embeddings into a ReLU MLP, and the Quantum Implicit Representation Network (QIREN). Our experiments demonstrate that QIREN can successfully reconstruct and extrapolate image structure based solely on coordinate-intensity pairs, showing competitive performance using a compact parameterized circuit with data reuploading. These results support the potential of quantum neural networks as viable function approximators for continuous visual data and open up possibilities for future work in higher-dimensional image tasks and low-data scenarios.

References

- [1] Z. H. Chao Chen, Yu-Shen Liu, “Sharpening neural implicit functions with frequency consolidation priors,” 2024. [Online]. Available: <https://arxiv.org/html/2412.19720v1>
- [2] G. W. Vincent Sitzmann, Michael Zollhöfer, “Scene representation networks: Continuous 3d-structure-aware neural scene representations,” 2020. [Online]. Available: <https://arxiv.org/abs/1906.01618>
- [3] J. S. R. N. S. L. Jeong Joon Park, Peter Florence, “Deep sdf: Learning continuous signed distance functions for shape representation,” 2019. [Online]. Available: <https://doi.org/10.48550/arXiv.1901.05103>
- [4] D. V. A. S. W. T. F. T. F. Kyle Genova, Forrester Cole, “Learning shape templates with structured implicit functions,” 2019. [Online]. Available: <https://doi.org/10.48550/arXiv.1904.06447>
- [5] M. T. P. H. R. M.-B. P. P. S. Jonathan T. Barron, Ben Mildenhall, “Mip-nerf: A multiscale representation for anti-aliasing neural radiance fields,” 2021. [Online]. Available: <https://doi.org/10.48550/arXiv.2103.13415>
- [6] J. Zhao, W. Qiao, P. Zhang, and H. Gao, “Quantum implicit neural representations,” 2024. [Online]. Available: <https://arxiv.org/abs/2406.03873>
- [7] N. G. Z. Y. Y. Q. K. Z. M.-H. H. P. R. D. T. Yuxuan Du, Xinbiao Wang, “Quantum machine learning: A hands-on tutorial for machine learning practitioners and researchers,” 2025. [Online]. Available: <https://doi.org/10.48550/arXiv.2502.01146>
- [8] E. F. Combarro, *Quantum Computing Foundations*. Cham: Springer International Publishing, 2022, pp. 1–24. [Online]. Available: https://doi.org/10.1007/978-3-031-05324-5_1
- [9] M. Schuld and F. Petruccione, *Machine Learning with Quantum Computers*, 2nd ed., ser. Quantum Science and Technology. Springer Cham, 2021, published: 17 October 2021, eBook edition. [Online]. Available: <https://doi.org/10.1007/978-3-030-83098-4>
- [10] M. Schuld, R. Sweke, and J. J. Meyer, “Effect of data encoding on the expressive power of variational quantum-machine-learning models,” *Phys. Rev. A*, vol. 103, p. 032430, Mar 2021. [Online]. Available: <https://link.aps.org/doi/10.1103/PhysRevA.103.032430>
- [11] A. Pérez-Salinas, A. Cervera-Lierta, E. Gil-Fuster, and J. I. Latorre, “Data re-uploading for a universal quantum classifier,” *Quantum*, vol. 4, p. 226, Feb. 2020. [Online]. Available: <https://doi.org/10.22331/q-2020-02-06-226>
- [12] B. R. Ali Rahimi, “Random features for large-scale kernel machines,” 2007.